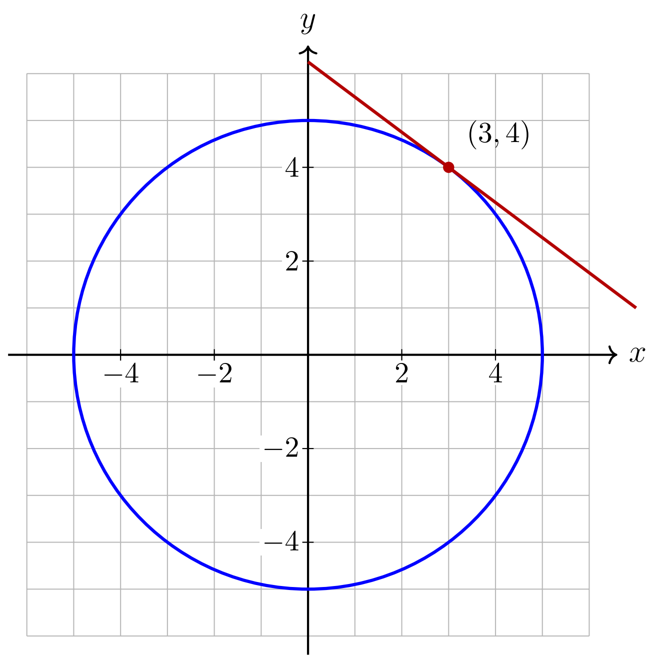
Differentiating Implicit Relations

Directions:

- Differentiate *both sides* with respect to x. Every time a y is differentiated, the chain rule attaches a factor $\dfrac{dy}{dx}$ — that factor is the whole method.
- Collect the $\frac{dy}{dx}$ terms on one side, factor it out, and divide. The answer usually involves both x and y, which is normal: a relation has different slopes at different points on the same vertical line.
- Then substitute the given point. Check first that the point really satisfies the relation.

Figure A — the circle $x^2 + y^2 = 25$ with its tangent line at the point $(3, 4)$. Problems 1 and 2 both find the slope of that tangent, first by solving for y and then implicitly.



1. Figure A. Near $(3, 4)$ the circle $x^2 + y^2 = 25$ is the graph of the upper branch $y = \sqrt{25 - x^2}$.

(a) Differentiate that branch directly to get $\dfrac{dy}{dx}$ in terms of x: ________

(b) Evaluate it at $x = 3$: ________

2. **Figure A.** Now do the same job implicitly. Differentiate $x^2 + y^2 = 25$ with respect to x, solve for $\frac{dy}{dx}$, and evaluate at the point $(3, 4)$. Your answer should match problem 1(b) — both methods give the slope of the same tangent line.

Answer: ________________

3. Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$ and evaluate it at the point $(1, 2)$. (Check that $(1, 2)$ is on the curve before you start; the xy term needs the product rule.)

Answer: ________________

4. Find $\frac{dy}{dx}$ for $x^3 + y^3 = 9$ and evaluate it at the point $(1, 2)$.

Answer: ________________

5. The parabola $y^2 = 4x$ passes through $(1, 2)$, and near that point it is the graph of the upper branch $y = 2\sqrt{x}$.

(a) Differentiate the branch directly, in terms of x: _________

(b) Differentiate $y^2 = 4x$ implicitly instead, and evaluate the result at $(1, 2)$: _________

6. Find $\dfrac{dy}{dx}$ for $x^2y + y^2 = 12$ and evaluate it at the point $(2, 2)$. (The first term is a product of two functions of x.)

Answer: __________________

7. Find $\dfrac{dy}{dx}$ for $x^2y^3 = 8$ and evaluate it at the point $(1, 2)$.

Answer: __________________

8. Challenge. The folium $x^3 + y^3 = 6xy$ passes through $(3, 3)$.

(a) Find $\dfrac{dy}{dx}$ implicitly and evaluate it at $(3, 3)$: _________

(b) Explain how the symmetry of the relation in x and y predicts that slope, without differentiating anything.